nature cities

Article

https://doi.org/10.1038/s44284-026-00533-w

Urban scaling and the dynamics of individual cities

Received: 1 April 2026

Accepted: 16 September 2026

Published online: 26 October 2026

Check for updates

Ulysse Marquis [1,2] & Marc Barthelemy [3,4,5]

Cities are often compared through scaling laws that relate population size to aggregate urban quantities such as infrastructure, socioeconomic activity or environmental impacts. These laws are influential because the scaling exponent describing how an urban quantity varies with population is often interpreted as indicating what happens when a city grows, with implications for urban theory, planning and policy. Here we show that this interpretation is generally misleading. A scaling relationship obtained by comparing many cities at one point in time does not, in general, describe the trajectory of any individual city. Instead, it reflects a statistical pattern across cities with different histories, constraints, institutions and growth paths. Relationships across cities can, therefore, suggest that an urban quantity increases either more slowly or more rapidly than population, even when individual cities follow much simpler dynamics, as we show for the relation between urban area and population. Cross-sectional scaling laws should, therefore, not be used to infer growth mechanisms or guide policy for individual cities: they describe statistical relationships across cities, rather than the temporal dynamics of individual cities.

Most cities are growing, and this growth affects many urban quantities at once: built area, housing, roads, transport systems, services, emissions, wages, innovation, crime, health and many others. A central quantitative question is, therefore, how these quantities change as city population increases. To address this issue, the urban scaling framework was introduced[1–9]. In its simplest form, it relates an aggregate urban quantity Y to population P, used as a proxy for city size[10], through

$$Y \approx P^{\beta}. \tag{1}$$

In practice, the exponent β is usually estimated from datasets containing many cities observed at a fixed time. The resulting relation is, therefore, cross-sectional: it describes an ensemble of cities at one moment, not the temporal evolution of a single city.

This framework has generated a large literature reporting apparent scaling laws for urban heat islands[11], economic productivity[12,13], crime[14,15], infrastructure networks[7,16], innovation[17,18], pollution and CO_2 emissions[19–23], building heights[24], road casualties[25] and depression rates[26]. Three regimes are usually distinguished. When $\beta < 1$, the per-capita value of Y decreases with city size, which is often interpreted as an economy of scale. When $\beta > 1$, Y grows more than proportionally with population, suggesting increased returns or amplification effects. The case $\beta = 1$ corresponds to linear scaling.

These interpretations matter because scaling exponents are often treated not only as descriptive quantities but also as indicators of mechanisms. For example, sublinear scaling of CO_2 emissions can be read as evidence that larger or denser cities are intrinsically more carbon-efficient[20,21,23], whereas the superlinear scaling of innovation or wages is often taken as evidence for agglomeration benefits. More generally, urban scaling exponents are frequently interpreted as revealing generic mechanisms of urban growth, efficiency and increasing returns.

This interpretation is deeply problematic. First, studies (for example, equation (1)) are shaped by data availability: cross-sectional snapshots of systems of cities are common, whereas long time series for individual cities are scarcer. This imbalance has encouraged the view that scaling exponents capture a fundamental property of urban

[1]Fondazione Bruno Kessler, Povo (TN), Italy. [2]Department of Mathematics, University of Trento, Povo (TN), Italy. [3]Université Paris-Saclay, CNRS, CEA, Institut de Physique Théorique, Gif-sur-Yvette, France. [4]Centre d'Analyse et de Mathématique Sociales (CNRS/EHESS) Paris, Paris, France. [5]Complexity Science Hub, Vienna, Austria. e-mail: ulyssepierre.marquis@unitn.it; marc.barthelemy@ipht.fr

systems, although their connection to individual trajectories is far from clear[27–30]. Second, the measured exponent depends on how cities are defined and how data are aggregated[31,32]. Third, small deviations from unity do not necessarily imply nonlinear urban dynamics[33–35].

The central question is, therefore, this: what does the exponent β measured across cities actually mean for an individual city? Previous work has already exposed this difficulty. For congestion-induced delays, aggregating data across cities and years yields an apparent nonlinear scaling, whereas individual city trajectories display strong path dependence and no common law[36]. Other studies have shown clear mismatches between cross-sectional regularities and city-level dynamics[27,29]. Ribeiro et al.[29] found city-specific longitudinal exponents for Brazilian cities, whereas Bettencourt et al.[30] emphasized that temporal exponents can become very large when urban quantities contain strong temporal trends or are better described by forms other than power laws (Supplementary Information, section I for additional discussions on the data in refs. 29,30). Similar conclusions arise from microlevel wage data, where superlinear cross-sectional scaling coexists with highly heterogeneous city trajectories[37]. Historical path dependence and technological change further complicate the interpretation of relations measured across systems of cities[38,39].

Pumain[28] argued that giving a dynamical meaning to cross-sectional scaling implicitly assumes a form of ergodicity in the systems of cities. In this context, ergodicity means identifying the temporal evolution of one city with the ensemble pattern observed across many cities at one date. This requires the strong assumption that cities have comparable histories and can be viewed, at least approximately, as scaled-up versions of one another.

These observations point to a fundamental issue: equation (1) does not imply that cities evolve along the same trajectories in the (Y, P) plane. It only constrains their relative positions at a given time. Even if every city follows a well-defined relation between Y and P over time, the exponent measured across cities need not equal the growth exponent of any individual city. Rather, as we show below, it is an emergent statistical quantity shaped by heterogeneous trajectories, historical contingencies, intercity correlations and the distribution of city sizes.

In this paper, we ask what the exponent measured across cities tells us about urban growth. We show that, in general, it does not describe the trajectory of any individual city. Instead, it combines how cities change over time, how they differ from one another and how city size correlates with city-specific characteristics. This decomposition identifies the restrictive conditions under which a scaling exponent can be interpreted as a genuine law of urban growth. More generally, urban scaling relations should be understood as statistical patterns emerging from a diverse system of cities, not as rules describing the growth of individual cities.

Results

We compare two perspectives on urban change: following the same cities through time, and comparing many cities at one date. These perspectives do not necessarily agree. Here we start from individual trajectories and interpret the cross-sectional pattern as the outcome of many heterogeneous histories observed in a single snapshot. The main question, therefore, is this: what information does a cross-sectional exponent provide about cities and their evolution? A more detailed discussion of previous approaches, and of the distinction between temporal and transversal scaling, is provided in Supplementary Information, section I.

Cross-sectional versus longitudinal scaling

We first clarify the distinction between transversal or cross-sectional scaling and longitudinal dynamics. Consider N cities at time t, with populations $P_1(t)\ldots P_N(t)$ and urban quantities $Y_1(t)\ldots Y_N(t)$. Cross-sectional, or transversal, scaling describes how Y varies across cities at fixed t:

$$Y_i(t) \approx Y_0(t)\, P_i(t)^{\beta_T(t)}, \quad \forall i,\ \text{fixed } t \tag{2}$$

where $\beta_T(t)$ is the transversal exponent and $Y_0(t)$ is a time-dependent prefactor. In log–log coordinates, $\beta_T(t)$ is usually estimated as the slope of a regression across cities at fixed time.

By contrast, longitudinal dynamics describe the evolution of Y_i within one city i as its population changes. In full generality, we get

$$Y_i(t) = F_i\,(P_i(t), t) \quad \forall t,\ \text{fixed } i, \tag{3}$$

where F_i may differ across cities and evolve in time. A power-law trajectory of the form

$$Y_i(t) = \mathcal{C}_i\, P_i(t)^{\beta_i} \quad \forall t,\ \text{fixed } i, \tag{4}$$

defines a city-specific exponent β_i and prefactor $\mathcal{C}_i$, but this is only a special case. In general, neither the functions F_i nor the exponents β_i need to be identical across cities.

From individual cities to urban scaling

We now ask how the transversal exponent emerges from individual trajectories. Let $x_i = \ln[P_i(t)]$ and $y_i = \ln[Y_i(t)]$. At fixed time, ordinary least squares (OLS) gives

$$\beta_T = \frac{\mathrm{Cov}(x, y)}{\mathrm{Var}(x)}, \tag{5}$$

with $\mathrm{Cov}(x, y) = N^{-1}\sum_{i=1}^{N}(x_i - \langle x\rangle)(y_i - \langle y\rangle)$, and $\langle\cdot\rangle$ is an average over cities (Supplementary Information, section II). This definition is purely cross-sectional and makes no assumption about the temporal path of any city. Other estimators could be used, but OLS is common in empirical studies and makes the contributions to β_T explicit; alternative fits give similar estimates for the cases studied below (Supplementary Information).

To connect β_T to individual trajectories, we write the local relation

$$y_i(t) \simeq \alpha_i(t) + \beta_i(t) x_i(t), \tag{6}$$

where $\beta_i(t)$ is the local longitudinal elasticity. It is estimated by a rolling regression of $\ln[Y_i]$ against $\ln[P_i]$ along city i, using a short window centered on t—typically 5 years in the data below—and approximates

$$\beta_i(t) = \frac{d\ln[Y_i(t)]}{d\ln[P_i(t)]}.$$

The intercept $\alpha_i(t)$ gives the local level of the trajectory in the $(\ln[P], \ln[Y])$ plane. Substituting equation (6) into equation (5) gives

$$\beta_T = \langle\beta\rangle + \frac{\mathrm{Cov}(x, \alpha)}{\mathrm{Var}(x)} + \frac{\mathrm{Cov}\,(x, (\beta - \langle\beta\rangle)x)}{\mathrm{Var}(x)}, \tag{7}$$

where all quantities are evaluated at the same date t, omitted for readability (Supplementary Information). Thus, β_T equals the mean longitudinal elasticity only if the two covariance terms vanish or cancel. Otherwise, it is an ensemble-level quantity shaped by heterogeneity and correlations across cities, not the dynamical exponent of individual trajectories.

Empirical illustrations

We test this decomposition (equation (7)) with three examples. The first is the "fundamental allometry" between urbanized area and population[40], where individual dynamics are approximately linear or piecewise linear. The second is wages, where trajectories are more heterogeneous. Congestion delay, previously studied in ref. 36, is revisited in the Supplementary Information. For each city, we estimate $\alpha_i(t)$ and $\beta_i(t)$, compute the different terms appearing in equation (7), and compare the predicted β_T with the exponent measured directly from cross-sectional fits.

https://doi.org/10.1038/s44284-026-00533-w

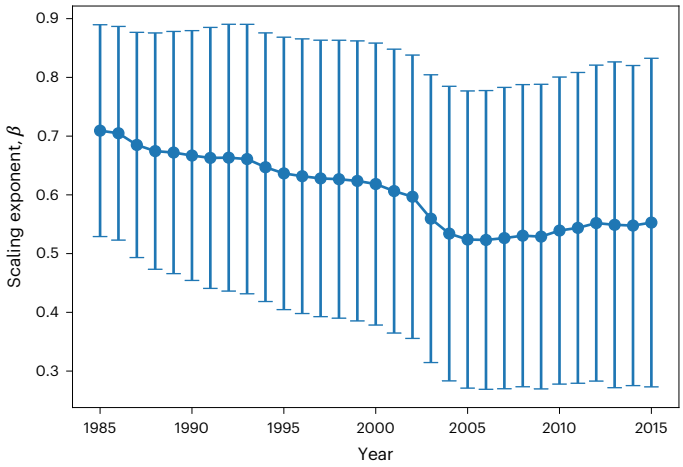


Fig. 1 | Cross-sectional scaling: area–population. Temporal evolution of β_T for the 19 cities studied in ref. 46, estimated by OLS in log–log space. Alternative estimators (Theil–Sen, weighted least squares) give similar results (Supplementary Information, section IV). Error bars indicate the exact 95% confidence interval of the OLS slope estimate, centered around the estimated exponent β_T, computed from the 19 cities.

Area–population relation. We first consider how built area A increases with population P, a classical urban scaling relation discussed in refs. 3,4,22,41–45. Early estimates gave $\beta_T \approx 3/4$ (ref. 41) and $\beta_T \approx 2/3$ (ref. 3). Batty and Ferguson[42] reported that most estimates are sublinear (Supplementary Information), whereas Burger et al.[45] found exponents near 5/6 for half of the countries studied and indistinguishable from one for the others.

Longitudinal measurements[46] use the 1800–2000 dataset[47] and the 1985–2015 datasets[48,49]. In all cases, individual cities are well described by

$$A_i(t) \approx a_i(t) P_i(t), \tag{8}$$

where $a_i(t)$ is the inverse density. The constant term is omitted because it does not affect variations. Trajectories can be strictly linear (a_i = constant) or piecewise linear, with $a_i = a_1$ before a city-specific time t^* and $a_i = a_2$ after it, usually with $a_2 < a_1$. Typical linear relations, yearly over 30 years and sporadically over about 200 years, are shown in Supplementary Fig. 4; density-breaking cases are discussed in ref. 46. Power-law fits $A \approx P^\beta$ are generally worse than linear or piecewise linear fits and yield dispersed exponents, $\beta = 1.06 \pm 0.36$, between 0.5 and 1.8 (Supplementary Fig. 5). Thus, population growth and spatial expansion are approximately linearly coupled along individual trajectories.

By contrast, cross-sectional fits for the same data give exponents different from one. For the data studied in ref. 46, we obtain Fig. 1 showing that $\beta_T \simeq 0.5$–0.7, robust to alternative estimators (Supplementary Information). Read literally, this would imply that density increases with city size, in contradiction with the approximately linear longitudinal dynamics. The difference arises because the inverse density a_i varies across cities and time, producing covariance terms in equation (7). Thus, nonlinear cross-sectional scaling can appear even when each city follows an essentially linear trajectory.

Figure 2 confirms this decomposition. The measured transversal exponent is accurately recovered by summing the mean longitudinal elasticity and the two covariance terms. The latter have opposite signs and do not fully cancel, shifting β_T away from the longitudinal value. The cross-sectional exponent, thus, reflects an aggregate statistical effect over the ensemble of cities rather than a genuine dynamical law governing the growth of individual cities.

To illustrate equation (7) analytically, we now consider a particularly simple case in which the longitudinal trajectories are piecewise linear. This assumption is introduced only to obtain an analytically tractable example and is not required for the general result (equation (7)).

Specifically, we show how a cross-sectional exponent different from unity can emerge even though the longitudinal dynamics of each city are locally linear. Motivated by the empirically observed piecewise linear trajectories, we consider a simple threshold model in which the area A_i of city i depends on its population P_i as

$$A_i = \begin{cases} a_1 P_i, & P_i < P_i^*, \\ a_2 P_i + (a_1 - a_2) P_i^*, & P_i \geq P_i^*. \end{cases} \tag{9}$$

The additive term ensures continuity at the city-specific threshold P_i^*, which is drawn from a given distribution. Each city grows linearly in each regime, but a transversal comparison mixes cities on different branches. Applying equation (7) to this case when all longitudinal exponents are $\beta_i = 1$ (Supplementary Information) gives

$$\beta_T = 1 + C(\ln[a_2] - \ln[a_1]), \tag{10}$$

where $C > 0$ depends on the distributions of P and P^*. Therefore, $\beta_T > 1$ if $a_2 > a_1$, and $\beta_T < 1$ if $a_2 < a_1$. Both the superlinear exponents observed in Angel et al.[47] and the sublinear exponents in ref. 46 can, thus, arise from linear longitudinal dynamics. Simulations of fixed and random thresholds (Supplementary Information, section III) confirm that $\beta_T = 1$ is recovered only when cities have identical dynamics and initial conditions; any source of heterogeneity—whether in initial populations, in thresholds or, more generally, in the dynamical parameters—creates correlations between α_i and x_i, causing β_T to depart from the true longitudinal exponent.

Wages. Total wages are a standard example of urban scaling. They have been reported to grow superlinearly with population[8], often interpreted as a consequence of enhanced social and economic interactions. In particular, interaction-based theories predict $\beta = 7/6 \simeq 1.17$ (ref. 50). For 363 US Metropolitan Statistical Areas in 2006, the transversal estimate is $\beta_T = 1.13 \pm 0.02$, close to this prediction. We use the full dataset of ref. 50, with yearly wages and population from 1969 to 2015.

Figure 3 shows that the cross-sectional exponent is remarkably stable over time, as for congestion delay, whereas individual trajectories are highly heterogeneous in slope and amplitude (Supplementary Fig. 6 shows the distribution of βi). Decomposing β_T with equation (7) shows that $\langle\beta\rangle \simeq 4$–$5$, far larger than both β_T and the predicted value of 7/6 (Supplementary Information, section V).

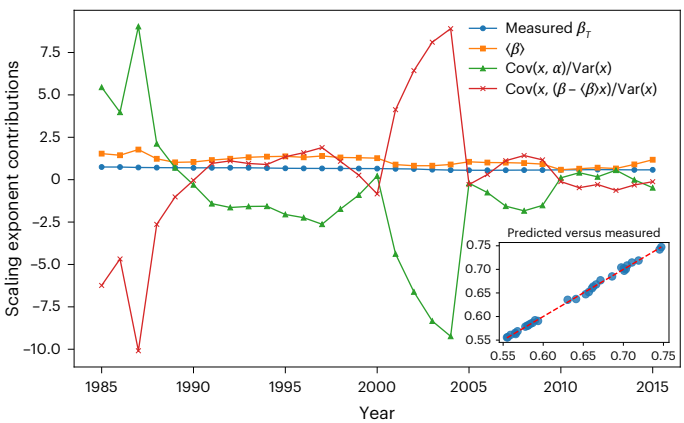


Fig. 2 | Statistical contributions to the transversal exponent. For the data discussed in Fig. 1, the local longitudinal exponent $\beta_i(t)$ is measured for each city. The average $\langle\beta\rangle$ is shown together with the two covariance contributions in equation (7). Their incomplete compensation explains the deviation between β_T and $\langle\beta\rangle$. Inset: measured exponent (obtained by directly fitting area versus population at fixed time) versus the prediction from equation (7).

https://doi.org/10.1038/s44284-026-00533-w

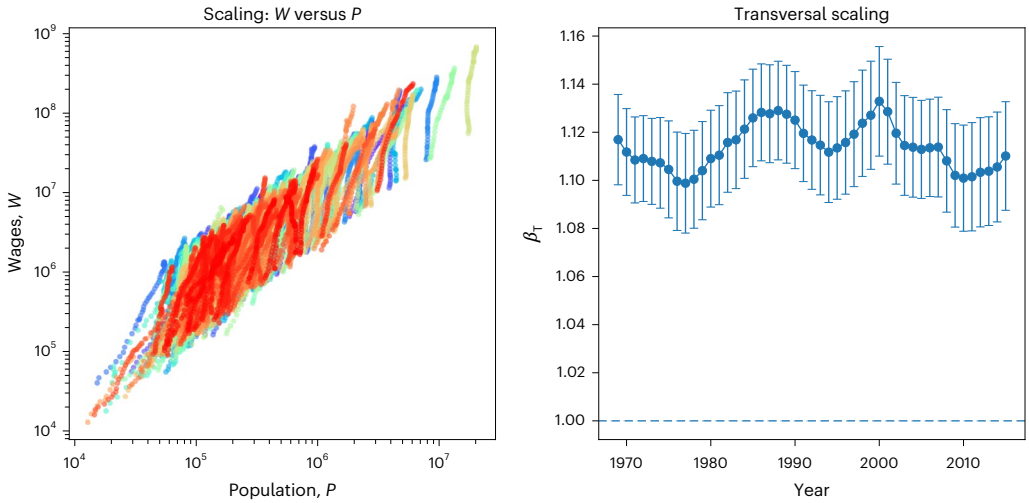


Fig. 3 | Wages: transversal scaling. Left: wages as a function of population for 363 US Metropolitan Statistical Areas observed yearly from 1969 to 2015. Contiguous colored points correspond to the same city; the color scale helps distinguish longitudinal growth from cross-sectional variation. Right: transversal exponents (β_T) as a function of time. The exponents were measured using OLS in log–log coordinates. The error bars, centered around these points, represent the exact 95% confidence intervals for the estimated OLS slope, calculated from the $n = 363$ cities. Data are obtained from ref. 50.

Individual cities, therefore, display much steeper wage trajectories than suggested by the cross-sectional fit. The stability of β_T results from a balance between heterogeneous longitudinal dynamics and cross-city correlations, not from a universal trajectory followed by all cities. The predicted transversal exponent reproduces the measured one (Supplementary Fig. 7), with remaining deviations attributable to the large dispersion of individual longitudinal exponents.

More generally, a transversal exponent can always be measured, whatever the temporal behavior of individual cities. Some cities may grow rapidly, others more slowly, and some may not even follow a power law. A cross-sectional fit will nevertheless return a value of β_T. This value measures a statistical property of the ensemble observed at a given time, not the longitudinal dynamics of any individual city.

Discussion

Urban scaling is often used to assess how city size affects quantities such as CO_2 emissions, crime, patents, infrastructure or economic output. The usual approach is to compare many cities at one date and fit how an aggregate quantity Y varies with population P. The exponent is then often interpreted as measuring the effect of population size: if it is larger than one, the per-capita quantity increases with city size; if it is smaller than one, the per-capita quantity decreases. This interpretation matters because it can support broad claims about cities. For instance, the sublinear scaling of CO_2 emissions has sometimes been taken to suggest that larger or denser cities are intrinsically more environmentally efficient, with possible implications for planning, density and governance.

The difficulty is that this interpretation relies on a strong assumption. Comparing different cities at a single date implicitly treats them as different stages of the same underlying trajectory. A city with population P_1 and value Y_1 is then interpreted as a scaled version of another city with population $P_2 > P_1$ and value Y_2. This would imply that as the first city grows from P_1 to P_2, it should approach a value of Y close to Y_2. Such a view assumes that cities follow the same dynamical path and differ mainly by size. This is very restrictive: cities have distinct histories, spatial constraints, institutions, technologies and planning trajectories.

Our results show that this assumption is generally not valid. The usual urban scaling exponent cannot, in general, be interpreted as a direct estimate of how an individual city changes as it grows. Rather, it is an aggregate quantity shaped by the distribution of city sizes, heterogeneity of city-specific trajectories, local responses at current population levels and correlations between these quantities. Cross-sectional scaling, therefore, reflects not only city-level dynamics but also the statistical structure of the urban system as a whole. The behavior encoded by the cross-sectional exponent is, in general, not followed by any individual city.

In some specific cases, such exponents may have a clear interpretation, namely, when larger systems are—to a good approximation—scaled-up versions of smaller ones, and when their present structure is not strongly determined by specific histories. A classical example is Kleiber's law[51–54], which relates metabolic rate to body mass through $B \approx M^{3/4}$. In that case, the exponent can be interpreted because organisms of different sizes share comparable biological constraints, particularly constraints associated with transporting energy and resources through the body.

For cities, the situation is different. Scaling exponents are sometimes used to support claims that larger cities are more innovative, more efficient or greener. Our results show that such conclusions should be drawn with caution. A sublinear or superlinear transversal exponent does not imply that increasing the population of a given city will produce the corresponding decrease or increase in the quantity per capita. It may instead reflect differences between cities: their histories, densities, infrastructures, economic structures, planning regimes or geographic constraints. Using cross-sectional scaling laws for prediction is especially problematic. For example, ref. 55 combines exogenous population-growth scenarios with cross-sectional scaling relations to forecast future greenhouse-gas emissions and traveled distances under different sprawling scenarios. Such approaches implicitly assume that scaling laws observed across cities remain valid along the future trajectories of individual cities, which can lead to misleading predictions and policy conclusions.

From a policy perspective, this calls for a shift in how urban scaling laws are interpreted. Cross-sectional scaling laws should neither be treated as fundamental laws of city growth nor used alone as guides for urban design or governance. They can reveal statistical regularities at the level of a system of cities, but they do not, by themselves, prescribe how a particular city will evolve or how it should be planned. Cross-sectional exponents are derived quantities: they emerge from the ensemble of longitudinal trajectories followed by individual cities. The fundamental objects are, therefore, these trajectories themselves—how emissions, infrastructure, land use, innovation, congestion or housing costs change as a given city grows. Understanding urban growth

requires starting from these temporal trajectories, because they contain the information needed to assess the effects of planning decisions, technological change, institutional constraints and path dependence.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

All data used in this paper are publicly and freely available from the sources cited in the manuscript. The area–population data are from the datasets described in refs. 47–49, and the wage data are from ref. 50. No restricted or proprietary data were used.

References

1. Zipf, G. K. *Human Behavior and the Principle of Least Effort* (Addison-Wesley Press, 1949).
2. Naroll, R. S. & von Bertalanffy, L. The principle of allometry in biology and social sciences. *Gen. Syst. Yearb.* **1**, 76–89 (1956).
3. Nordbeck, S. Urban allometric growth. *Geogr. Ann. B* **53**, 54–67 (1971).
4. Woldenberg, M. J. An allometric analysis of urban land use in the United States. *Ekistics* **36**, 282–290 (1973).
5. Longley, P. A., Batty, M. & Shepherd, J. The size, shape and dimension of urban settlements. *Trans. Inst. Br. Geogr.* **16**, 75–94 (1991).
6. Pumain, D. *Scaling Laws and Urban Systems*. Working Paper 2004-02-002 (Santa Fe Institute, 2004).
7. Kuhnert, C., Helbing, D. & West, G. B. Scaling laws in urban supply networks. *Phys. A* **363**, 96–103 (2006).
8. Bettencourt, L. M., Lobo, J., Helbing, D., Kühnert, C. & West, G. B. Growth, innovation, scaling, and the pace of life in cities. *Proc. Natl Acad. Sci. USA* **104**, 7301–7306 (2007).
9. Chen, Y. Characterizing growth and form of fractal cities with allometric scaling exponents. *Discrete Dyn. Nat. Soc.* **2010**, 194715 (2010).
10. Batty, M. The size, scale, and shape of cities. *Science* **319**, 769–771 https://doi.org/10.1126/science.1151419 (2008).
11. Oke, T. City size and the urban heat island. *Atmos. Environ.* **7**, 769–779 (1973).
12. Sveikauskas, L. The productivity of cities. *Q. J. Econ.* **89**, 393–413 (1975).
13. Pumain, D., Paulus, F., Vacchiani-Marcuzzo, C. & Lobo, J. An evolutionary theory for interpreting urban scaling laws. *Cybergeo* 10.4000/cybergeo.2519 (2006).
14. Glaeser, E. L. & Sacerdote, B. Why is there more crime in cities?. *J. Polit. Econ.* **107**, S225–S258 https://doi.org/10.1086/250109 (1999).
15. Puttock, S., Barros, U., Pinheiro, D. & Oliveira, M. Larger cities, more commuters, more crime? The role of inter-city commuting in the scaling of urban crime. *Crime Sci.* **14**, 19 (2025).
16. Samaniego, H. & Moses, M. E. Cities as organisms: allometric scaling of urban road networks. *J. Transp. Land Use* **1**, 21–39 (2008).
17. Bettencourt, L. M. A., Lobo, J. & Strumsky, D. Invention in the city: increasing returns to patenting as a scaling function of metropolitan size. *Res. Policy* **36**, 107–120 (2007).
18. Arbesman, S., Kleinberg, J. M. & Strogatz, S. H. Superlinear scaling for innovation in cities. *Phys. Rev. E* **79**, 016115 https://doi.org/10.1103/PhysRevE.79.016115 (2009).
19. Louf, R. & Barthelemy, M. Scaling: lost in the smog. *Environ. Plann. B* **41**, 767–769 https://doi.org/10.1068/b4105c (2014).
20. Glaeser, E. L. & Kahn, M. E. The greenness of cities: carbon dioxide emissions and urban development. *J. Urban Econ.* **67**, 404–418 (2010).
21. Oliveira, E. A., Andrade, J. S. & Makse, H. A. Large cities are less green. *Sci. Rep.* **4**, 4235 (2014).
22. Rybski, D. et al. Cities as nuclei of sustainability?. *Environ. Plann. B* **44**, 425–440 (2017).
23. Fragkias, M., Lobo, J., Strumsky, D. & Seto, K. C. Does size matter? Scaling of CO2 emissions and US urban areas. *PLoS ONE* **8**, e64727 (2013).
24. Schläpfer, M., Lee, J. & Bettencourt, L. M. A. Urban skylines: building heights and shapes as measures of city size. Preprint at https://doi.org/10.48550/arXiv.1512.00946 (2015).
25. Cabrera-Arnau, C., Prieto Curiel, R. & Bishop, S. R. Uncovering the behaviour of road accidents in urban areas. *R. Soc. Open Sci.* **7**, 191739 https://doi.org/10.1098/rsos.191739 (2020).
26. Stier, A. J. et al. Evidence and theory for lower rates of depression in larger us urban areas. *Proc. Natl Acad. Sci. USA* **118**, e2022472118 https://doi.org/10.1073/pnas.2022472118 (2021).
27. Xu, G., Zhou, Z., Jiao, L., Dong, T. & Li, R. Cross-sectional urban scaling fails in predicting temporal growth of cities. Preprint at https://doi.org/10.48550/arXiv.1910.06732 (2019).
28. Pumain, D. Dynamique des entités géographiques et lois d'échelle dans les systèmes complexes : la question de l'ergodicité. *Math. Sci. Hum.* **191**, 51–63 (2010).
29. Ribeiro, F. L., Meirelles, J., Netto, V. M., Neto, C. R. & Baronchelli, A. On the relation between transversal and longitudinal scaling in cities. *PLoS ONE* **15**, e0233003 (2020).
30. Bettencourt, L. M. A. et al. The interpretation of urban scaling analysis in time. *J. R. Soc. Interface* **17**, 20190846 (2020).
31. Louf, R. & Barthelemy, M. How congestion shapes cities: from mobility patterns to scaling. *Sci. Rep.* **4**, 5561 (2014).
32. Arcaute, E. et al. Constructing cities, deconstructing scaling laws. *J. R. Soc. Interface* **12**, 20140745 (2015).
33. Shalizi, C. R. Scaling and hierarchy in urban economies. Preprint at https://doi.org/10.48550/arXiv.1102.4101 (2011).
34. Leitao, J. C., Miotto, J. M., Gerlach, M. & Altmann, E. G. Is this scaling nonlinear?. *R. Soc. Open Sci.* **3**, 150649 (2016).
35. Barthélemy, M. Tomography of scaling. *J. R. Soc. Interface* **16**, 20190526 (2019).
36. Depersin, J. & Barthelemy, M. From global scaling to the dynamics of individual cities. *Proc. Natl Acad. Sci. USA* **115**, 2317–2322 (2018).
37. Keuschnigg, M. Scaling trajectories of cities. *Proc. Natl Acad. Sci. USA* **116**, 13 759–13 761 (2019).
38. Connor, D. S., Kemeny, T. & Storper, M. Frontier workers and the seedbeds of inequality and prosperity. *J. Econ. Geogr.* **24**, 393–414 (2024).
39. Connor, D. S. et al. Big cities fuel inequality within and across generations. *PNAS Nexus* **4**, pgae587 (2025).
40. Ribeiro, F. L. & Rybski, D. Mathematical models to explain the origin of urban scaling laws. *Phys. Rep.* **1012**, 1–39 (2023).
41. Stewart, J. Q. Suggested principles of "social physics". *Science* **106**, 179–180 (1947).
42. Batty, M. & Ferguson, P. Defining city size. *Environ. Plann. B* **38**, 753–756 (2011).
43. Ortman, S. G., Cabaniss, A. H. F., Sturm, J. O. & Bettencourt, L. M. A. The pre-history of urban scaling. *PLoS ONE* **9**, e87902 (2014).
44. Hamilton, M. J., Milne, B. T., Walker, R. S. & Brown, J. H. Nonlinear scaling of space use in human hunter–gatherers. *Proc. Natl Acad. Sci. USA* **104**, 4765–4769 (2007).
45. Burger, J. R. et al. Global city densities: re-examining urban scaling theory. Preprint at https://doi.org/10.48550/arXiv.2210.08067 (2022).
46. Marquis, U., Artime, O., Gallotti, R. & Barthelemy, M. Universal roughness and the dynamics of urban expansion. *Phys. Rev. Lett.* **135**, 187403 (2025).

47. Angel, S., Parent, J., Civco, D. L. & Blei, A. *Atlas of Urban Expansion—2012 Edition, Volume 1: Areas and Densities* (New York University, 2012).
48. DLR—German Aerospace Center. *World Settlement Footprint Evolution (WSF Evolution)* https://download.geoservice.dlr.de/WSF_EVO/ (2019).
49. World Population Review & Macrotrends. *Historical Population (1950–2025) for Urban Areas*. https://worldpopulationreview.com/ (2024).
50. Bettencourt, L. M. A. The origins of scaling in cities. *Science* **340**, 1438–1441 (2013).
51. Kleiber, M. Body size and metabolism. *Hilgardia* **6**, 315–353 (1932).
52. Kleiber, M. Body size and metabolic rate. *Physiol. Rev.* **27**, 511–541 (1947).
53. Enquist, B. J., Brown, J. H. & West, G. B. Allometric scaling of plant energetics and population density. *Nature* **395**, 163–165 (1998).
54. West, G. B., Brown, J. H. & Enquist, B. J. A general model for the origin of allometric scaling laws in biology. *Science* **276**, 122–126 (1997).
55. Hankey, S. & Marshall, J. D. Impacts of urban form on future us passenger-vehicle greenhouse gas emissions. *Energy Pol.* **38**, 4880–4887 (2010).

Acknowledgements

M.B. thanks the Complexity Science Hub in Vienna and R. Prieto-Curiel, as well as the University of Padua and Manlio De Domenico, for their invitations, for the opportunity to present and discuss this work, and for many interesting and stimulating discussions.

Author contributions

U.M. conceived the study, developed the analysis code and theoretical framework, and wrote the manuscript. M.B. conceived the study, developed the theoretical framework, performed the analysis and wrote the manuscript.

Funding

The authors received no specific funding for this work.

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s44284-026-00533-w.

Correspondence and requests for materials should be addressed to Ulysse Marquis or Marc Barthelemy.

Peer review information *Nature Cities* thanks Yuchao Chen, Fabiano Ribeiro and the other, anonymous, reviewer(s) for their contribution to the peer review of this work.

Reprints and permissions information is available at www.nature.com/reprints.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.

© The Author(s), under exclusive licence to Springer Nature America, Inc. 2026

nature portfolio

Corresponding author(s): Marc Barthelemy

Last updated by author(s): Sep 8, 2026

Reporting Summary

Nature Portfolio wishes to improve the reproducibility of the work that we publish. This form provides structure for consistency and transparency in reporting. For further information on Nature Portfolio policies, see our Editorial Policies and the Editorial Policy Checklist.

Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

n/a	Confirmed	
☐	☒	The exact sample size (*n*) for each experimental group/condition, given as a discrete number and unit of measurement
☐	☒	A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
☒	☐	The statistical test(s) used AND whether they are one- or two-sided *Only common tests should be described solely by name; describe more complex techniques in the Methods section.*
☒	☐	A description of all covariates tested
☒	☐	A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
☐	☒	A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
☒	☐	For null hypothesis testing, the test statistic (e.g. *F*, *t*, *r*) with confidence intervals, effect sizes, degrees of freedom and *P* value noted *Give P values as exact values whenever suitable.*
☒	☐	For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
☒	☐	For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
☒	☐	Estimates of effect sizes (e.g. Cohen's *d*, Pearson's *r*), indicating how they were calculated

Our web collection on statistics for biologists contains articles on many of the points above.

Software and code

Policy information about availability of computer code

Data collection	python3.13
Data analysis	python3.13

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio guidelines for submitting code & software for further information.

Data

Policy information about availability of data

All manuscripts must include a data availability statement. This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our policy

All data used in this paper are publicly and freely available from the sources cited in the manuscript. The area–population data are from the datasets described in [47–49], and the wage data are from Ref. [50]. No restricted or proprietary data were used.

Research involving human participants, their data, or biological material

Policy information about studies with human participants or human data. See also policy information about sex, gender (identity/presentation), and sexual orientation and race, ethnicity and racism.

Reporting on sex and gender	n/a
Reporting on race, ethnicity, or other socially relevant groupings	n/a
Population characteristics	n/a
Recruitment	n/a
Ethics oversight	n/a

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☐ Life sciences ☐ Behavioural & social sciences ☒ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see nature.com/documents/nr-reporting-summary-flat.pdf

Ecological, evolutionary & environmental sciences study design

All studies must disclose on these points even when the disclosure is negative.

Study description	*Briefly describe the study. For quantitative data include treatment factors and interactions, design structure (e.g. factorial, nested, hierarchical), nature and number of experimental units and replicates.*
Research sample	*Describe the research sample (e.g. a group of tagged Passer domesticus, all Stenocereus thurberi within Organ Pipe Cactus National Monument), and provide a rationale for the sample choice. When relevant, describe the organism taxa, source, sex, age range and any manipulations. State what population the sample is meant to represent when applicable. For studies involving existing datasets, describe the data and its source.*
Sampling strategy	*Note the sampling procedure. Describe the statistical methods that were used to predetermine sample size OR if no sample-size calculation was performed, describe how sample sizes were chosen and provide a rationale for why these sample sizes are sufficient.*
Data collection	*Describe the data collection procedure, including who recorded the data and how.*
Timing and spatial scale	*Indicate the start and stop dates of data collection, noting the frequency and periodicity of sampling and providing a rationale for these choices. If there is a gap between collection periods, state the dates for each sample cohort. Specify the spatial scale from which the data are taken*
Data exclusions	*If no data were excluded from the analyses, state so OR if data were excluded, describe the exclusions and the rationale behind them, indicating whether exclusion criteria were pre-established.*
Reproducibility	*Describe the measures taken to verify the reproducibility of experimental findings. For each experiment, note whether any attempts to repeat the experiment failed OR state that all attempts to repeat the experiment were successful.*
Randomization	*Describe how samples/organisms/participants were allocated into groups. If allocation was not random, describe how covariates were controlled. If this is not relevant to your study, explain why.*
Blinding	*Describe the extent of blinding used during data acquisition and analysis. If blinding was not possible, describe why OR explain why blinding was not relevant to your study.*

Did the study involve field work? ☐ Yes ☒ No

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☒	☐ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Plants

Seed stocks	n/a
Novel plant genotypes	n/a
Authentication	n/a